

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 17)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

Table 1 shows the annual emissions of two pollutants in the UK in 1970 and 2015.

Table 1

Pollutant	Emissions in 1970 / thousand tonnes	Emissions in 2015 / thousand tonnes	Main source in 2015
Sulfur dioxide	6 400	220	Energy industries
Nitrogen oxides	2 700	930	Road transport

0 1

1

Calculate the percentage reduction in sulfur dioxide emissions between 1970 and 2015.

[2 marks]

Answer

%

0 1

2

Suggest **three** reasons why sulfur dioxide emissions fell by a much larger percentage than nitrogen oxide emissions.

[3 marks]

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0 1

3

Explain why a reduction in NO_x emissions would also reduce concentrations of tropospheric ozone and PANs.

[2 marks]

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0 1 . 4

Explain why lichens are used to monitor sulfur dioxide pollution.

[3 marks]

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10

0 2 . 1

Explain what is meant by the **embodied energy** of a vehicle and how it can be reduced.

[3 marks]

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0 2 . 2

Describe **four** principles of designing vehicles for end of life.

[4 marks]

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0 2 . 3

Explain why there is an optimum speed for fuel efficiency in a car.

[2 marks]

0 2 . 4

Give **one** way driver behaviour or vehicle systems can reduce fuel use in congested traffic.

[1 mark]

10

0 3

A student used 0.5 m x 0.5 m grid quadrats to survey ground flora in two meadows, one grazed and one ungrazed. Table 1 shows the results for the grazed meadow (10 quadrats).

Table 1

Species	Number of individuals
Ribwort plantain	48
Perennial ryegrass	120
White clover	36
Daisy	24
Creeping buttercup	12

0 3 . 1

Explain why a 0.5 m x 0.5 m quadrat was chosen rather than a 10 cm x 10 cm or 100 m x 100 m quadrat.

[2 marks]

0 3 . 2

Calculate Simpson's Index of Biodiversity (D) for the grazed meadow using $D = N(N-1) / \sum n(n-1)$. Show your working.

[3 marks]

Answer

03 . 3

The ungrazed meadow had $D = 1.8$ with the same number of species. Explain what this shows.

[2 marks]

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03 . 4

Explain why a point quadrat would be better than a frame quadrat in tall vegetation.

[2 marks]

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03 . 5

Give **one** limitation of using the DAFOR abundance scale.

[1 mark]

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10

0 4 . 1

Describe the two processes that drive the North Atlantic Conveyor.

[2 marks]

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0 4 . 2

Explain how global climate change could weaken the North Atlantic Conveyor and describe the possible consequence for the climate of the UK.

[4 marks]

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0 4 . 3

Explain why the effects of climate change on UK temperatures are difficult to predict.

[3 marks]

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0 4 . 4

Describe how deep ocean currents are monitored.

[1 mark]

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10

05 . 1

Explain how the control of emission **location** and emission **timing** can reduce the severity of pollution. Use examples.

[6 marks]

Emission location

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Emission timing

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05 . 2

Explain the **polluter pays** principle and suggest why it encourages pollution prevention.

[2 marks]

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05 . 3

Explain the **precautionary principle** as applied to the release of a new chemical.

[2 marks]

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0 6 . 1

Which of the following water sources is usually the most convenient for large communities? Tick one box.

[1 mark]

- A Desalinated seawater
- B River water
- C Rainwater catchment
- D Newly discovered aquifers

☐
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0 6 . 2

Explain why the quality requirements for water differ between uses, using **two** named uses.

[2 marks]

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0 6 . 3

Explain why inter-basin transfers may be used and give **one** disadvantage.

[2 marks]

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5

0 7 . 1

Explain the difference between nuclear fission and nuclear fusion.

[2 marks]

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0 7 . 2

State the sources of the two fuels used in fusion research.

[2 marks]

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1

2

07 . 3

Explain why very high temperatures, a vacuum and a magnetic field are needed in a toroidal (torus) fusion reactor.

[3 marks]

07 . 4

Give **two** potential advantages of fusion over fission and one reason why it is not yet a commercial energy source.

[3 marks]

10

0 8

The Aral Sea in central Asia was once the fourth largest lake in the world. From the 1960s water from the two rivers feeding it (the Amu Darya and Syr Darya) was diverted to irrigate cotton and rice. Table 1 shows changes in the lake.

Table 1

Year	1960	1980	2000	2015
Lake area / km ²	68 000	52 000	24 000	7 000
Salinity / g dm ⁻³	10	18	45	over 100
Commercial fish catch / thousand tonnes	40	10	0	0

0 8**1**

Calculate the percentage reduction in the area of the Aral Sea between 1960 and 2015.

[2 marks]**Answer**

%**0 8****2**

Explain why the salinity of the lake increased as the area fell.

[2 marks]

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0 8**3**

Explain why the fish catch fell to zero, and describe **two** other environmental consequences of the shrinking of the lake.

[4 marks]

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08 . 4

Suggest why disagreements over the rivers continue between the countries of the region.

[2 marks]

10

09

Table 1 shows features of three rechargeable battery types.

Table 1

Battery type	Energy density / Wh kg ⁻¹	Charge-discharge cycles	Storage efficiency / %	Cost per kWh / £	Safety issues
Lead-acid	35	500	80	100	Toxic lead, acid
Nickel-cadmium	50	1500	75	250	Toxic cadmium
Lithium-ion	200	2000	92	150	Fire risk

09 . 1

Use Table 1 to explain why lithium-ion batteries are used in electric vehicles but lead-acid batteries are still used for stationary storage in some off-grid solar systems.

[4 marks]

09 . 2

Give **two** other factors that affect the viability of a rechargeable battery.**[2 marks]**

1

2

09 . 3

Describe how Vehicle-to-Grid (V2G) systems could reduce the need for peaking power stations.

[2 marks]

09 . 4

Explain how Power-to-Gas (P2G) systems use existing infrastructure.

[2 marks]

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10 . 1

Explain how organic wastes and inorganic nutrients cause water pollution and evaluate the effectiveness of the methods used to control them.

[9 marks]

Do not write outside the box

Extra space

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10 . **2** Give **one** reason why algae cause eutrophication problems more quickly than rooted plants when nutrients are added to a lake.

[1 mark]

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Write an essay on one of the following topics.

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Discuss the extent to which the study of ozone depletion illustrates how scientific investigation can lead to the successful solution of an environmental problem.

[25 marks]

OR

1 1 . 2

Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

[illegible]

[illegible]

END OF QUESTIONS